Clifford Geometry and Local-Unitary Invariants of Six-Qudit AME Tensors

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Abstract

This preparation draft organizes exact results on six-qudit, minimal-support absolutely maximally entangled stabilizer tensors arising from six-point projective arcs. The present proved package classifies local-Clifford equivalence in a non-Reed–Solomon pencil by one projective scalar, detects a sharp change in the logical Clifford group at the Reed–Solomon locus, and constructs exact local-unitary invariants that separate the distinguished non-Reed–Solomon tensors from Reed–Solomon tensors. We prove that every local-unitary intertwiner between equal-phase CSS states of linear $[6, 3, 4]_q$ MDS codes is local Clifford. On the admitted odd non-Reed–Solomon pencil, local-unitary equivalence, local-Clifford equivalence, and equality of the projective scalar therefore coincide.

1 Introduction

A six-point arc in the projective plane determines a length-six dimension-three MDS code and hence an equal-phase CSS stabilizer tensor. For odd local dimension this gives a concrete family of minimal-support AME$(6, q)$ states in which projective geometry, coding equivalence, and local quantum equivalence can be compared exactly.

This draft separates three questions. First, which tensors are equivalent under local Clifford operations? Second, which geometric distinctions survive arbitrary local unitaries? Third, when do the local-unitary and local-Clifford orbit partitions agree inside the non-GRS pencil? The key rigidity theorem answers the third question at the stronger level of all equal-phase CSS states from linear $[6, 3, 4]_q$ MDS codes: every local-unitary intertwiner is local Clifford.

2 From six-arcs to AME tensors

This section will fix the projective, code-equivalence, stabilizer, and party-permutation conventions. It will state the six-arc/MDS/CSS/AME dictionary used by every later theorem and identify the degenerate and GRS boundary loci.

3 Local-Clifford classification of the non-GRS pencil

This section will define the admitted parameter pencil, its projective coordinate z, and the theorem equating projective, monomial-code, and local-Clifford equivalence with equality of z.

4 Local-unitary invariants

This section will develop the marginal-moment formula and the exact copy-contraction invariants. It will distinguish proved separation theorems from the open completeness question for arbitrary local-unitary orbits.

5 The logical-Clifford phase

This section will compare the fixed-party logical symplectic kernels on the GRS and non-GRS loci and explain the resulting operational distinction in each encoder view.

6 The four-copy transport sheaf

This section will replace the raw contraction matrix by the reduced transport operator, derive the exceptional divisor, and explain the associated orbit multiplicities. Any use of this operator in a complete local-unitary classification must first certify which coefficients are arbitrary-LU invariants.

7 Verification and scope

This section will state the trust boundary for conceptual proofs, exact symbolic calculations, finite certificates, independent replays, and any future formalization. The final version will cite the paper-local evidence manifest and give exact replay commands.